Surface electromyograms

How to work as a group

This lab involves three parts. First, assembling your lab equipment, taking measurements and writing a report.

As usual, you only need to turn in one report. But feel free to have fun and each collect your own data and compare the readings with each other. If there are substantial discrepancies, feel free to put the graphs for multiple people in the same report (you don't have to tell me whose pictures are whose unless you want to).

Building the board

The main work in this lab is building your equipment. This lab uses the equipment to collect a *surface EMG* (sEMG), which is a reading (at the skin surface) of the electrical voltage that your body's neurons use to drive your muscles. Later, we will use similar equipment to take an *ECG*, which measures the electrical voltages that your heart produces.

There are separate documents on how to build the equipment and how to use a scope; you can find both on the course web page. There is also a safety document (which you must read) and a legal document (which you must read, sign and hand in) before doing the lab.

Where to attach the skin pads

For this lab, you should attach:

- The blue electrode is the positive input. Attach it right at the middle of your biceps muscle (the belly of the muscle, which is the largest-diameter part of the muscle).
- The black electrode is the negative input. Attach it slightly above your elbow, trying to ensure that it is not on top of any muscle (bone or fat is fine).
- The red electrode is called RLD, and we'll talk about it more in the ECG lab. Attach it to your wrist, on the same arm as the other electrodes.
- we have skin-prep tape available. It is essentially very fine sandpaper; if you scrape your skin *lightly* with it (don't scrape yourself red!) then you get a better connection and stronger signals. Most people do fine without it until cold weather comes and your skin becomes drier.

For all three electrodes, as usual, if you can avoid locations with a lot of hair you'll get better skin contact – as well has having a less painful time removing the pad later.

What to measure

You should get three or four readings.

1. With your arm completely relaxed and your lower arm resting on a table.
2. Put your arm back on the table, but now "make a muscle" – clenching your biceps as hard as you can.
3. Gradual increase in force from your arm quite relaxed to a maximal contraction.
4. If you have time (i.e., if you don't have to spend most of the lab figuring out how to use the scopes!), we have plenty of AD8232 boards, and our scopes have two channels. You can thus probe two different muscles at the same time; e.g., biceps and

triceps. Put yourself in the place of an amputee who would like to, e.g., use their biceps and triceps sEMG to independently control their artificial hand (e.g., use their biceps to make their wrist bend and use their triceps to pinch their thumb and forefinger together). Can you operate these muscles independently at will?

Discussion questions

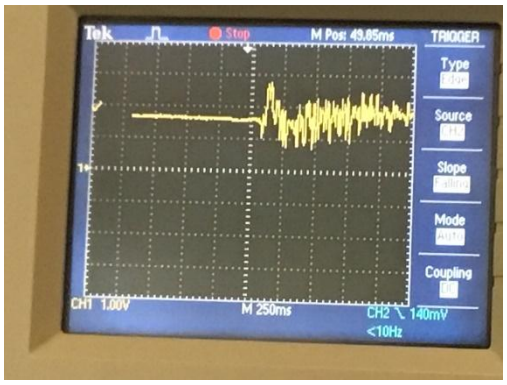
1. Hopefully your brain has learned by this point in your life how to create a smoothly-increasing force in your biceps muscle! What physiological factors might still make it be difficult to create a smoothly-increasing ramp as measured by an sEMG signal? (20 points)
2. How does interlacing make the measurement of a smooth sEMG gradation possible? Note that we're not asking how it enables your brain to create a smooth force ramp-up in your muscles, but rather how it enables seeing a smoothly-increasing sEMG in response to the ramp. (20 points)
3. An artificial-hand prosthetic might have two movements: thumb/forefinger gripping and wrist rotating. You've shown that it's easy to read an sEMG from the biceps muscle; in fact, the triceps is also easy to access. For an amputee who has lost their hand (i.e., a mid-radial amputee), what might be pros and cons of using the biceps & triceps sEMGs to control a prosthetic hand's gripping and wrist rotation respectively? Try to give one "pro" and two "cons." (30 points)

What to turn in

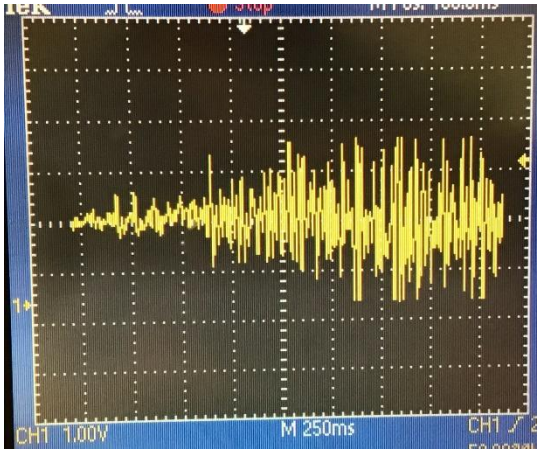
- Turn in your sEMG pictures (30 points), along with the answers to the three questions above.
- You only need to turn in one lab report per group (though, again, you're encouraged to all do the experiment).

Sample waveforms

Here are some waveforms that I collected. They can give you an idea what to look for. And if you don't want to turn in your own readings (which do constitute protected medical records), feel free to use these as your own.



Biceps totally relaxed on the left, and maximally contracted on the right.



Gradual increase in contraction strength.